MILITARY SPECIFICATION

GRINDING MACHINES, CARBIDE TOOL BIT AND CARBIDE TOOL BIT AND CHIP BREAKER COMBINATION, ELECTRIC MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers electric motor driven grinding machines to use for wet or dry grinding of carbide tool bits with equipment and accessories specified herein.

1.2 Classification. This specification covers grinding machines of the following types, classes, styles, and sizes. The type, class, and size required for a specific procurement shall be specified (see 6.2.1).

Type I - Roughing and Finishing

Class 1 - Single spindle, double end wheel mount
Class 2 - Double spindle, single end wheel mount
Class 3 - Single spindle, single end wheel mount

Style A - Wet
Style B - Dry
Style C - Wet and Dry

Size	Grinding wheel (outside dimension)
6	6-inch
10	10-inch
14	14-inch

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3415

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Type II - Combination Chipbreaker and Finishing

Style A - Wet
Style B - Dry
Style C - Wet and Dry

Size	Grinding wheel (outside dimension)
6	6-inch

Type III - Single Straight Wheel Tool Grinder

Style A - Wet
Style B - Dry
Style C - Wet and Dry

Size	Grinding wheel (outside dimension)
14	14-inch
16	16-inch
20	20-inch

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part Of this specification to the extent specified herein.

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U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NEMA MG 1 - Motors and Generators (DoD adopted).

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for (DoD adopted).

ANSI B7.1 - Abrasive Wheels, Use, Care and Protection of, Safety Requirements for the (DoD adopted).

ANSI B11.9 - Safety Requirements for the Construction, Care and Use of Grinding Machines (DoD adopted).

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives (DoD adopted).

ANSI B74.2 - Shapes and Sizes of Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels, Specifications for (DoD adopted).

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for (DoD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN GEAR MANUFACTURERS ASSOCIATION

AGMS 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Myer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO-54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

ISO-R1122 - Gears, Glossary of, Geometrical Definitions.

(Application for copies should be addressed to the International Organization for Standardization, c/o The American National Standards Institute, ATTN : Sales Department, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The grinding machine shall be new and one of the manufacturer's current models capable of hand grinding carbide tool bits to various angles in accordance with the requirements herein. Type 11 machines shall also be capable of grinding chipbreaker grooves. Style A machines shall be capable of wet grinding; style B machines shall be furnished for dry grinding; and style C machines shall be designed for wet or dry grinding. Al 1 machines shall be capable of grinding to the same tolerances and finishes based on operator control. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements,

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dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The grinding machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the grinding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and OSHA 2206 that are applicable to the machine itself. In the event of a conflict between the requirements of ANSI B11.9 and OSHA 2206, the requirements of OSHA 2206 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The grinding machine shall not contain mercury or mercury compounds nor be exposted to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

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3.2.8 Environmental protection. The machine shall be designed and constructed cuch that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of the installation.

3.2,9 Lubrication. A lubrication system shall be provided to insure adequate lubrication for all moving parts. The system shall deliver lubrication oil to all mating gears, sliding parts, and bearings (except the sealed-for-lilfe and self-lubricating type) . Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit Installation of replacement parts without modification to the part or the machine.

3.3 Construction. The grinding machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part .

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt. fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

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3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Gears. All gears and pinions used in the machine or its components, shall be properly selected for the intended application and fabricated to either English (USA) standards per AGMA 390.03 and AGMA 360.02 or Metric (SI) standards per the International Organization for Standardization, ISO-1122, ISO-54, and FED-STGD-376.

3.4 Components. The machine shall consist basically of the following components. Any additional components necessary for the operation of the machine shall be furnished in accordance with manufacturer's standard practice.

3.4.1 Base. The base shall be a casting of iron or iron alloy of box-like construction. The base shall support or house the motor drive(s), the grinding wheel assembly(s), worktable(s), and all other associated parts. The coolant system and its related components shall be housed within the base. Door(s) or readily removable cover(s) shall be provided for servicing the internally housed components. The base shall have means for securing the machine to a mounting surface or foundation.

3.4.2 Grinding wheel assembly. The grinding wheel assembly shall include the work spindle(s) and related components, spindle housing(s), drive mechanism(s), grinding wheel flanges, and grinding wheels. Work spindles shall be hardened and ground, mounted on preloaded ball or roller bearings and be dynamically balanced. Work spindles shall support grinding wheels that run true within 0.0005 inch TIR (total indicator reading) both radially and axially. Spindle and motor bearings shall be protected from dirt, coolant, and other foreign matter by seals. Spindle drives shall utilize either the motor shaft as a work spindle or a belt and pulley. Unless otherwise specified (see 6.2.1), motors shall be of the reversing type. Motor control shall be through a switch having forward, stop and reverse (when applicable). Type I, class 1 machines shall be fitted with a grinding wheel on each end of a single spindle. Type I, class 2 machines shall have two grinding wheel assemblies, each having an individual work spindle and driving mechanism and each capable of working as an independent unit. Type I, class 3 machines shall have a single spindle fitted with a single grinding wheel. Type II machines shall have a single spindle fitted with a grinding wheel on one end of the spindle and a chipbreaker on the other end. Type III machines shall be fitted with a straight grinding wheel on one end of a single spindle, designed with a safety interlock and two spindle speeds to maintain optimum peripheral speed throughout wheel life. Unless otherwise specified (see 6.2.1), each work spindle on type I and type II machines

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shall be fitted with a cup type silicone carbide grinding wheel and type III machines shall be fitted with a straight silicone carbide grinding wheel. Style A and style C machines shall include a coolant reservoir, pump unit, coolant and sludge pans, coolant spouts with control valves, and spray guards. Style B machines shall have wheelhoods equipped with exhaust outlets. When specified (see 6.2.1), style B machines shall have a drip pot and tray for wet grinding.

3.4.3 Worktable. Type I, class 1 and class 2 machines and type II machines shall be furnished with two worktables; type I, class 3 machines and type III machines shall be furnished with a single worktable. Worktables shall be of the dimensions given in table I. The surface of all worktables shall be finished for a work mounting surface. The table shall have coolant troughs or similar means for draining spent coolant into splash pans . Worktables shall provide tool support for face grinding on type I machines and the grinding wheel side of type 11 machines. Tool support for peripheral grinding shall be provided on the chip breaker side of type 11 machines and on type 111 machines.

3.4.3.1 Table tilt. Each worktable, except on the chipbreaker wheel side of type II machines, shall be designed to tilt no less than 15 degrees above the horizontal and 25 degrees below the horizontal. Tilting worktable shall be provided with means of locking the table after positioning. Table tilt shall be indicated on an accurately graduated, tilt-angle scale. Tilt angle scale markings shall be in one degree increments with numerical characters indicated at zero and every ten degrees. Angle indication accuracy of the scale shall be within 30 minutes at all points through the table tilt range. Provisions shall be made for adjustment of the tables or the grinding wheels to compensate for wheel wear.

3.4.3.2 Table movement. Type II machines shall be provided with a reciprocating worktable on the chipbreaker wheel side. The table shall be capable of reciprocating longitudinally and traversely with vertical adjustment. . Cross slide feed shall be by a screw and dial on dovetail ways. The table shall traverse on ball bearing ways with an adjustable lever. A micrometer barrel and dial, calibrated to 0.001 inch (see 3.3.7), shall provide sensitive vertical feed during grinding.

3.4.3.3 Protractor gage groove. Each worktable, except for the chipbreaker wheel side on type II machines, shall be grooved longitudinally to accommodate a protractor gage. The groove shall extend the entire length of the table and shall be aligned with wheel faces within 0.001 inch per inch.

3.4.4 Wheelhods. A wheelhood or guard shall be provided for each grinding wheel. The hood shall guard the grinding wheel and shall be located in such a manner that it will not interfere with wheel changing. The hood shall be in accordance with ANSI B7.1.

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3.4.5 Hand brake. Unless otherwise specified (see 6.2.1), a hand brake, positioned convenient to the operator and operable from both wheel positions, shall provide quick stopping of the wheel(s). The brake shall stop the wheel(s) from maximum revolutions per minute to a dead stop within 10 seconds.

3.4.6 Protractor. A protractor shall be furnished with each machine. Gage graduation shall be such that it is zeroed with the body at a 90 degree angle to the protractor gage groove. The protractor gage shall be adjustable 45 degrees both directions from zero. Scale markings shall be increments of not less than one degree with the numerical value indicated at zero and each ten degree increment. Angle indication accuracy of the scale shall be within 15 minutes throughout the entire protractor range. When required (see 6.2.1), an extension plate on the protractor gage body shall be provided for angle grinding of short tools.

3.4.7 Work light. A 120 volt work light shall be provided on each machine. The work light shall have a protective shield and shall be attached to the machine in such a manner as to allow the light to be swung over either wheel. When more than one light is required, the number and location shall be as specified (see 6.2.1).

3.4.8 Coolant and sludge pans. Coolant and sludge pans shall be provided for style A and C machines located in such a manner as to extend beyond the worktable to catch coolant and sludge from grinding operations. Facilities shall be provided for draining the pans without their removal from the machine.

3.4.9 Wheel dresser. A wheel dressing fixture, equipped with a dressing tool tipped with a commercial diamond of not less than 1/2 carat in size, shall be furnished with each grinding machine. The fixture shall be guided by the protractor gage groove for alignment with the face of the wheel, and shall be equipped with a feed screw to enable the operator to take light uniform truing cuts.

3.4.10 Vise. Each type 11 machine shall have one vise fitted to the worktable on the chipbreaker side of the machine. The vise shall have angular and rotary adjustment graduated in degrees and shall be capable of off-center grinding. The vise shall have a maximum tool width capacity of at least 2-1/2 inches.

3.4.11 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1.

3.4.11.1 Mounting plates. Mounting plates for the cup grinding wheels on type I and type II machines shall be provided. Unless otherwise specified (see 6.2.1), mounting plates shall be of the following dimensions:

a, For 6-inch cup wheels: 6-inch diameter mount ing plate; 1-1/4-inch diameter centerhole; and four (4), 9/32-inch diameter holes on a 3-1/4-inch bolt circle (B.C).

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b. For 10-inch cup wheels: 10-inch diameter mounting plate; 2-inch diameter centerhole; and six (6), 13/32-inch diameter mounting holes on a 3-1/2-inch B.C.

c. For 14-inch cup wheels: 14-inch diameter mounting plate; 1-1/4-inch diameter centerhole; and four (4), 17/32-inch diameter mounting holes on 8 inch B.C.

A backing plate shall also be supplied for the chipbreaker wheel side of type 11 machines. Unless otherwise specified (see 6.2.1), the backing plate for a 6-inch diamond chipbreaker wheel shall be a minimum of 3-7/8 inches 6 inches in diameter with a 1-1/4-inch diameter centerhole.

3.4.12 Coolant system. A coolant system shall be provided on type I, style A and style C machines, type II machines, and type III, style A and style C machines. It shall be a flood type system and shall include a sump or reservoir, a power pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to control the direction and amount of coolant flow.

3.4.13 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.4. 13.1 Motors . Motors shall be rated for continuous duty and shall have a ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure in accordance to ANSI/NEMA MG-1.

3.4.14 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. Graduated table tilt scales and locks.

c. Adjustable splash guards.

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d. Grinding wheel mounting plates.

e. Protractor.

f. Adjustable work light(s).

g. Wheel dresser.

3.4.15 Optional equipment. Optional equipment shall be furnished as specified (see 6.2.1).

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Accuracy and performance. The grinding machine shall be capable of grinding hand-positioned carbide tool bits and chipbreakers as required in normal shop practice. The hand brake, worktable surface, worktable tilt and protractor shall be within the required limits when tested in accordance with 4.6.

3.7 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.8 Marking of instruction plates, indicating plates, and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.8.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.8.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specifed (see 6.2.1), the nameplate shall contain the information listed below.

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TABLE I. Sizes and capacities.

CHARACTERISTICS	REQUIREMENTS												
	Type I									Type II		Type III	
	Class 1			Class 2			Class 3						
	Style A, B, C												
	Size 6	Size 10	Size 14	Size 6	Size 10	Size 14	Size 6	Size 10	Size 14	Size 6	Size 14	Size 16	Size 20
Wheel size, inches dia x face x W x thk x bor.	6 x 1 x 1-1/4 x 1-1/4	10 x 1-1/2 x 1-1/2 x 2	14 x 1-1/2 x 4 x 1-1/4	6 x 1-1/2 x 2 x 1-1/4	10 x 1-1/2 x 2 x 2	14 x 1-1/2 x 4 x 1-1/4	6 x 1-1/2 x 2 x 1-1/4	10 x 1-1/2 x 2 x 2	14 x 1-1/2 x 4 x 1-1/4	6 x 1 x 1-1/4 x 1-1/4	14 x 2 - x 1/1/4	16 x 2-1/2 - x 6	20 x 2-1/2 - x 9
Wheel speed, surface feet per minute	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500	5500
Spindle speed, RPM	3600	1900	1350	3270	1970	1400	3270	1970	1400	3450	1635 - 2180	1550-1980	1240 - 1525
Spindle motor, HP	1/2	2	3	1-1/2	1-1/2	3	1	1	2	1	2	3	3
Number of toolrest tables	2	2	2	2	2	2	1	1	1	2	1	1	1
Table size, inches	10x6-3/4	19 x 10	22 x 12	19 x 10	19 x 10	24 x 12	19 x 10	19 x 10	24 x 12	14 x 8-1/2	10 x 10	10 x 10	10 x 10
Table tilt, degrees from horizontal													
Chipbreaker side													
Table traverse, inches	—	—	—	—	—	—	—	—	—	8	--	--	--
Wheel size, inches Dia x face x hole	—	—	—	—	—	—	—	—	—	4-6 x 1/8 - 1/2 x 1-1/4	--	--	--
Wheel speed, surface feet per minute	—	—	—	—	—	—	—	—	—	5500	--	--	--
Vise capacity, inches	—	—	—	—	—	—	—	—	—	2-1/2	--	--	--
Coolant tank capacity, Gallons	12	15	15	12	15	15	12	15	15	12	15	15	15

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in the above table. When the largest size and capacity machine shown in the above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is given, the required performance is from the stated minimum or less to the stated maximum or greater.

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designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.10 Workmanship. Workmanship of the grinder and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5, the tests in 4.6 and

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the Inspection 4.7. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined for design, construction, materials, components , electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The grinder shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms , and accessories shall be verified during the trial period.

4.6.2 Performance tests. Each grinding machine shall be subjected to the following tests:

4.6.2.1 Hand brake test. When a hand brake is supplied on the machine (see 3.4.5), the machine shall be allowed to run with no load for at least five minutes. The switch shall then be turned to the “off” position and the hand brake applied simultaneously. Application of the hand brake shall bring the wheels to a dead stop with 10 seconds.

4.6.2.2 Worktable surface test. The surface of each worktable shall be tested in five places on each side of the protractor gage groove. The tests shall be made with a precision straight edge and a 0.001-inch feeler gage. Straight edge shall be true within 0.0001 inch. The table shall be flat within 0.001 inch, concave only.

4.6.2.3 Worktable tilt test. A protractor which is accurate within one minute shall be used to determine the angle of the worktable in relation to the flat surface of the grinding wheel. The table angle shall be determined at the zero marking and at the two extremes of angle adjustment. Error shall not exceed 30 minutes at any point tested.

4.6.2.4 Protractor gage test. The protractor gage shall be removed from the worktable and shall be tested for accuracy at the zero setting and at the 15, 30 and 45 degree settings each way from zero, using a protractor which is accurate within one minute. Error shall not exceed 15 minutes at any point tested.

4.6.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.7.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section S.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing, and special marking shall be accomplished in accordance with MIL-M-18058.

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6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in any toolroom or production shop where off-hand carbide tool bit grinding is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c* When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI)< state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Reversing type motor, if not required (see 3.4.2).

j. Grinding wheels, if different (see 3.4.2).

k. Drip pot and tray on style B machines, if required (see 3.4.2).

l. Hand brake, if not required (see 3.4.5).

m. Protractor extension plate, if required (see 3.4.6).

n. Number and location of work light(s), if different (see 3.4.7).

o. Mounting or backing plates, if different (see 3.4.11.1).

p. Electrical system and input power supply, if different (see 3.4.13).

q. Motor enclosure, if different (see 3.4.13.1).

r. Standard equipment, if different (see 3.4.14).

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s. Optional equipment, fully described, if required (see 3.4.15).

t. Size and capacities, if different (see 3.5).

u. Electromagnetic interference control, if required (see 3.7); specify Class and Requirement per MIL-STD-461.

v. Lubrication chart or plate, if different (see 3.8.1).

w. Nameplate, if different (see 3.8.2).

x. First article inspection, if different (see 4.3).

y" Quality conformance inspection, if different (see 4.4).

z. Specify special packaging, packing, and marking, if required (see 5.1).

6.2.2 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.l(g).

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6.5 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-G-80025C:

Type I, class 3	New
Type 111, Size 14	New
Size 16	New
Size 20	New

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force 99

Review Activities:
Army - AL
Navy - SH
Air Force - None
DLA-GS

User Activities:
Army - None
Navy - OS, AS
Air Force - 84

Preparing Activity:
DLA - IP

Project Number:
3415-0152

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-G-80025C	Grinding Machines, Carbide Tool Bit and Carbide Tool Bit and Chip Breaker Combination, Electric Motor Driven

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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